Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building and upgrading business systems, automation workflows, and data-driven applications that improve operational efficiency for internal users and customer-facing platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built a multi-tenant education platform on **Azure SQL** and Blob Storage for **500+ users**, reducing dashboard latency by **71%** (**1.2s to 350ms**) through query and indexing improvements.
- Implemented automated data and job-processing workflows with **FastAPI, Supabase, and SQL Server**, processing **5K+ monthly jobs** with **3x** higher throughput and stable production execution.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Built file-ingestion and grading workflows for **10K+ requests/day** using **Flask, MySQL, AWS RDS, and REST APIs**, supporting batch processing of **1K+** files with reliable validation and monitoring.
- Created automation around rubric generation, OCR, and evaluation workflows for **150+ instructors**, reducing feedback turnaround time by **60%** while improving operational consistency during peak grading loads.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with a **microservices architecture**, automating VM builds and **20+** unit/integration test suites using **Jenkins, PyTest, and Mockito** across Windows, Linux, and macOS.
- Troubleshot and optimized API integrations with **AWS Glue, S3, and OOP patterns**, improving response times by **30%**, pipeline efficiency by **68%**, and lowering technical debt by **20%**.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a **resource management platform** with **MERN**, WebSockets, and REST APIs, synchronizing **2M+ records** across locations and supporting **8** business workflows for allocation, billing, and notifications.
- Automated builds, testing, and deployment with **Jenkins, Docker, and Azure SQL**, reducing resource wastage by **42%**, increasing lead conversions by **15%**, and improving release validation to **95%+**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask** APIs and Firebase-backed booking workflows for a cross-platform service application handling **50K+ records**, improving confirmation speed by **40%** and reducing manual refund handling by **70%**.
- Designed pricing, coupon, payment, and analytics workflows with **Databricks, Spark, and webhooks**, supporting sub-**200ms** booking APIs and hourly reporting for operational decisions.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++
Backend & Systems: Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, Microservices, OOP
Databases & Platforms: PostgreSQL, MySQL, MongoDB, Azure SQL, SQL Server, Firebase, AWS, Microsoft Azure
Testing & Delivery: PyTest, Mockito, Jenkins, GitHub Actions, Docker, CI/CD, Debugging, Monitoring
Engineering: Data Structures, Algorithms, Workflow Automation, ETL Pipelines, System Troubleshooting, Agile

Projects

Chat2SQL GitHub | *Java, TypeScript, Docker, Maven*
- Built an AI-powered SQL development workspace on top of the Chat2DB project, supporting **10+** databases and accelerating query authoring, schema exploration, and data management from a unified interface.

CPU-scheduling Simulator Live | *C++, React.js, EJ-2*
- Implemented FCFS, SJF, SRT, RR, priority, and MLQ algorithms with interactive visualization, reducing debugging time by **20%** and demonstrating up to **25%** lower average turnaround time across **1K+** simulations.

HMIS – Hospital Management Information System Live | *ABDM APIs, Appwrite, Twilio, TypeScript*
- Developed a workflow-driven healthcare platform supporting **100+ daily appointments**, reducing patient record retrieval by **2x** and cutting administrative workload by **30%** through consent-aware records and automated notifications.